

CSC6033 E#1 – Matrix Multiplication and Dot Product Steps

Example a)

Matrix A (2×2):

2 5

3 8

Matrix B (2×4):

6 9 11 13

12 18 22 26

Dot Product Steps:

Row 1 of A: [2 5]

- $[2\ 5] \times [6\ 12] = 2 \times 6 + 5 \times 12 = 12 + 60 = \mathbf{72}$
- $[2\ 5] \times [9\ 18] = 2 \times 9 + 5 \times 18 = 18 + 90 = \mathbf{108}$
- $[2\ 5] \times [11\ 22] = 2 \times 11 + 5 \times 22 = 22 + 110 = \mathbf{132}$
- $[2\ 5] \times [13\ 26] = 2 \times 13 + 5 \times 26 = 26 + 130 = \mathbf{156}$

Row 2 of A: [3 8]

- $[3\ 8] \times [6\ 12] = 3 \times 6 + 8 \times 12 = 18 + 96 = \mathbf{114}$
- $[3\ 8] \times [9\ 18] = 3 \times 9 + 8 \times 18 = 27 + 144 = \mathbf{171}$
- $[3\ 8] \times [11\ 22] = 3 \times 11 + 8 \times 22 = 33 + 176 = \mathbf{209}$
- $[3\ 8] \times [13\ 26] = 3 \times 13 + 8 \times 26 = 39 + 208 = \mathbf{247}$

Resulting Matrix:

72 108 132 156

114 171 209 247

Example b)

Matrix A (3×3):

1 2 4

8 16 32

64 128 256

Matrix B (3×2):

3 5

7 9

11 13

Dot Product Steps:

Row 1 of A: [1 2 4]

- $[1\ 2\ 4] \times [3\ 7\ 11] = 1 \times 3 + 2 \times 7 + 4 \times 11 = 3 + 14 + 44 = \mathbf{61}$
- $[1\ 2\ 4] \times [5\ 9\ 13] = 1 \times 5 + 2 \times 9 + 4 \times 13 = 5 + 18 + 52 = \mathbf{75}$

Row 2 of A: [8 16 32]

- $[8\ 16\ 32] \times [3\ 7\ 11] = 8 \times 3 + 16 \times 7 + 32 \times 11 = 24 + 112 + 352 = \mathbf{488}$
- $[8\ 16\ 32] \times [5\ 9\ 13] = 8 \times 5 + 16 \times 9 + 32 \times 13 = 40 + 144 + 416 = \mathbf{600}$

Row 3 of A: [64 128 256]

- $[64\ 128\ 256] \times [3\ 7\ 11] = 64 \times 3 + 128 \times 7 + 256 \times 11 = 192 + 896 + 2816 = \mathbf{3904}$
- $[64\ 128\ 256] \times [5\ 9\ 13] = 64 \times 5 + 128 \times 9 + 256 \times 13 = 320 + 1152 + 3328 = \mathbf{4800}$

Resulting Matrix:

61 75

488 600

3904 4800